IMAT 모의고사 6회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> The human gut contains trillions of microbes, collectively called the microbiota. Observational studies have linked differences in gut microbes to obesity, depression and autoimmune disease. Yet association is not causation: illness, diet and medication can all reshape the microbiota, so altered microbes may be a consequence of disease rather than its cause. The strongest evidence of causation comes from experiments in which microbes transferred from sick humans into germ-free mice produce similar symptoms in the animals.

1. According to the passage, why do observational studies alone fail to prove that microbes cause disease?

A. Disease and its treatment may themselves change the microbiota.
B. The microbiota is too small to study.
C. Obesity is not a disease.
D. Observational studies are always fraudulent.
E. Mice do not have gut microbes.

2. The experiments with germ-free mice are considered stronger evidence because they:

A. introduce the microbes first and then observe whether symptoms follow.
B. measure diet only.
C. remove all microbes from humans.
D. are cheaper.
E. use larger samples of humans.

3. Which Italian scientist is known for pioneering work on nerve growth factor, for which she shared the 1986 Nobel Prize in Physiology or Medicine?

A. Fabiola Gianotti
B. Margherita Hack
C. Maria Montessori
D. Samantha Cristoforetti
E. Rita Levi-Montalcini

4. The Universal Declaration of Human Rights was adopted by the UN General Assembly in:

A. 1957
B. 1966
C. 1948
D. 1919
E. 1945

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. Some antibiotics are effective against viruses — this claim is false. Which of the following must therefore be true?

A. Some viruses are antibiotics.
B. Some antibiotics are not effective against bacteria.
C. All viruses resist antibiotics and bacteria.
D. All antibiotics are effective against viruses.
E. No antibiotics are effective against viruses.

6. A shop increases prices by 20%, and then a further 10%. What is the overall percentage increase?

A. 32%
B. 31%
C. 30%
D. 28%
E. 33%

7. In a hospital, 3 doctors can see 36 patients in 4 hours. At the same rate, how many patients can 5 doctors see in 6 hours?

A. 108
B. 90
C. 80
D. 60
E. 72

8. What is the missing number? 1, 1, 2, 3, 5, 8, 13, __

A. 18
B. 20
C. 26
D. 21
E. 24

9. A runner completes laps of a 400 m track at 80 s per lap. A second runner completes laps at 100 s per lap. They start together. After how many seconds will the faster runner first lap (overtake by one full lap) the slower runner?

A. 200 s
B. 800 s
C. 320 s
D. 1000 s
E. 400 s

Section 3 — Biology (Q10–32)

10. Which cytoskeletal component forms the spindle fibres during cell division?

A. Keratin
B. Collagen fibres
C. Microtubules
D. Intermediate filaments
E. Actin filaments

11. Which statement about the genetic code is correct?

A. The code is degenerate: most amino acids are coded by more than one codon.
B. The code differs completely between humans and bacteria.
C. There are 20 possible codons.
D. Each amino acid is coded by exactly one codon.
E. Codons consist of two nucleotides.

12. Apoptosis is:

A. the fusion of gametes.
B. uncontrolled cell death caused by injury.
C. a form of mitosis.
D. programmed cell death.
E. cell division in bacteria.

13. Which molecule enters the Krebs cycle by combining with oxaloacetate?

A. Lactate
B. Pyruvate
C. NADH
D. Glucose
E. Acetyl-CoA

14. A woman carrier of an X-linked recessive disease and an unaffected man have a daughter. What is the probability that the daughter is a carrier?

A. 1/2
B. 3/4
C. 1/4
D. 0
E. 1

15. In photosynthesis, the oxygen released comes from:

A. ATP
B. glucose
C. chlorophyll
D. water
E. CO_2

16. Which part of the brain is primarily responsible for coordination of balance and fine movement?

A. Thalamus
B. Cerebellum
C. Medulla oblongata
D. Cerebrum
E. Hypothalamus

17. Which of the following describes a phospholipid?

A. A chain of glucose monomers
B. Four fused carbon rings
C. A polymer of amino acids
D. Glycerol, two fatty acids and a phosphate group with a hydrophilic head
E. Three fatty acids attached to glycerol

18. Saltatory conduction occurs because action potentials:

A. travel only in unmyelinated axons.
B. travel backwards along the axon.
C. are produced in the dendrites.
D. are conducted by Schwann cells.
E. jump between nodes of Ranvier in myelinated axons.

19. Which blood component is most important for clot formation?

A. Lymphocytes
B. Plasma water
C. Albumin
D. Platelets
E. Erythrocytes

20. A man of blood group A (heterozygous) and a woman of blood group B (heterozygous) have children. What is the probability that a child has blood group O?

A. 0
B. 1/4
C. 1/2
D. 1/8
E. 3/4

21. The hypothalamus controls the anterior pituitary mainly by:

A. nerve impulses directly to the anterior pituitary.
B. secreting insulin.
C. releasing hormones carried in the hypophyseal portal system.
D. producing ADH stored in the anterior pituitary.
E. producing thyroxine.

22. How many hydrogen bonds hold together a DNA fragment with 10 base pairs, of which 4 are G–C?

A. 20
B. 34
C. 24
D. 30
E. 26

23. Which statement about antibodies is correct?

A. All antibodies bind all antigens.
B. Antibodies are produced by red blood cells.
C. Antibodies consist of two heavy and two light chains.
D. Each antibody has four identical polypeptide chains.
E. Antibodies are carbohydrates.

24. The countercurrent multiplier in the kidney allows:

A. increased blood pressure.
B. secretion of glucose.
C. synthesis of urea.
D. filtration of red blood cells.
E. production of concentrated urine.

25. In which phase of meiosis does crossing over occur?

A. Anaphase II
B. Metaphase I
C. Anaphase I
D. Prophase II
E. Prophase I

26. Which enzyme property explains why it works best at a particular pH?

A. pH changes increase activation energy permanently.
B. pH changes the enzyme into a carbohydrate.
C. pH has no effect on enzymes.
D. pH changes alter the ionisation of amino acids in the active site.
E. pH changes alter the primary structure.

27. Which of the following is a sex-chromosome aneuploidy?

A. Turner syndrome (45,X)
B. Cri-du-chat syndrome
C. Patau syndrome
D. Down syndrome
E. Edwards syndrome

28. Which process moves large particles into a cell by engulfing them in a vesicle?

A. Exocytosis
B. Active transport through a carrier
C. Osmosis
D. Facilitated diffusion
E. Phagocytosis

29. Retroviruses such as HIV use which enzyme to convert their RNA into DNA?

A. Ligase
B. DNA polymerase III
C. RNA polymerase II
D. Reverse transcriptase
E. Integrase only

30. The hormone secretin stimulates the pancreas to secrete:
A. insulin.
B. bile.
C. glucagon.
D. pepsinogen.
E. bicarbonate-rich fluid.

31. Which of the following is the correct order in protein synthesis and secretion?
A. Nucleus → lysosome → Golgi → membrane
B. Smooth ER → nucleus → ribosome → membrane
C. Ribosome → mitochondria → Golgi → membrane
D. Ribosome on rough ER → ER lumen → vesicle → Golgi → secretory vesicle → plasma membrane
E. Golgi → ribosome → rough ER → vesicle → membrane

32. Which of the following provides evidence for evolution from molecular biology?
A. Fossil record
B. Geographical distribution of species
C. Vestigial organs
D. Artificial selection
E. Similar DNA and protein sequences among related species

Section 4 — Chemistry (Q33–47)

33. How many moles of atoms in total are present in 1 mol of glucose, $C_6H_{12}O_6$?
A. 12
B. 24
C. 180
D. 18
E. 6

34. What is the oxidation number of nitrogen in NH_4^+?
A. +3
B. −3
C. −4
D. +4
E. +5

35. A solution of pH 4 is diluted 100 times with water. What is the new pH (approximately)?
A. 5
B. 6
C. 4
D. 2
E. 8

36. Which of the following is a noble gas?

A. N
B. Ne
C. F
D. O
E. Na

37. Which molecule has a trigonal planar shape?

A. NH_3
B. PCl_3
C. CH_4
D. BF_3
E. H_2O

38. How many litres of O_2 at STP (22.4 L/mol) are needed to burn 1 mol of CH_4 completely?

A. 67.2 L
B. 44.8 L
C. 89.6 L
D. 22.4 L
E. 11.2 L

39. What is the product of the hydration of ethene (in the presence of an acid catalyst)?

A. Ethanoic acid
B. Ethyl ethanoate
C. Ethanol
D. Ethanal
E. Ethane

40. 100 mL of 0.5 mol/L NaCl is mixed with 400 mL of water. What is the final concentration?

A. 0.2 mol/L
B. 0.05 mol/L
C. 0.1 mol/L
D. 0.125 mol/L
E. 0.25 mol/L

41. Which of the following molecules can form hydrogen bonds with each other?

A. CH_4
B. CH_3OCH_3
C. CH_3OH
D. CO_2
E. CCl_4

42. Which of the following salts gives a basic solution in water?

A. NH_4Cl
B. NaCl
C. KNO_3
D. CH_3COONa
E. Na_2SO_4

43. For the reaction A + B → C, the rate doubles when [A] doubles and quadruples when [B] doubles. What is the overall order?

A. 4
B. 1
C. 0
D. 3
E. 2

44. Which functional group is present in an amide?

A. $–CONH_2$
B. –COOH
C. –OH
D. $–NH_2$ only
E. –CN

45. Which quantum number determines the shape of an orbital?

A. Magnetic (m_l)
B. Mass number
C. Principal (n)
D. Azimuthal / angular momentum (l)
E. Spin (m_s)

46. In the electrolysis of molten NaCl, what is produced at the cathode?

A. Cl_2
B. O_2
C. Na
D. H_2
E. NaOH

47. Which compound is an isomer of ethanol?

A. Ethanal
B. Dimethyl ether
C. Methanol
D. Propanol
E. Ethane

Section 5 — Physics & Mathematics (Q48–60)

48. A car decelerates uniformly from 30 m/s to rest over 75 m. What is the magnitude of its deceleration?

A. 0.4 m/s^2
B. 6 m/s^2
C. 4 m/s^2
D. 12 m/s^2
E. 2 m/s^2

49. A 10 kg object is pushed up a frictionless incline to a height of 2 m. If 250 J of work is done, how much kinetic energy does it have at the top? ($g = 10\ m/s^2$)

A. 0 J
B. 250 J
C. 50 J
D. 450 J
E. 200 J

50. A 1.5 V cell is connected to a 3 Ω resistor. How much energy is transferred in 20 s?

A. 15 J
B. 1.5 J
C. 0.15 J
D. 30 J
E. 90 J

51. A pendulum has a period of 2 s. If its length is quadrupled, what is the new period?

A. 16 s
B. 8 s
C. 4 s
D. 1 s
E. 2 s

52. An object of density 800 kg/m^3 is placed in water (1000 kg/m^3). What fraction of its volume is submerged when it floats?

A. 0.2
B. 1.25
C. 1.0
D. 0.5
E. 0.8

53. A 3 kg object is in free fall. What is its weight? ($g = 10\ m/s^2$)

A. 30 N
B. 0 N
C. 10 N
D. 3 N
E. 300 N

54. What is the frequency of a wave whose period is 0.02 s?

A. 2 Hz
B. 50 Hz
C. 500 Hz
D. 0.02 Hz
E. 20 Hz

55. Simplify: log(50) + log(2) (base 10).

A. 100
B. 2
C. log 52
D. 1
E. 25

56. In triangle ABC, angle A = 90°, AB = 5 and BC = 13. What is AC?

A. 10
B. 14
C. 8
D. √194
E. 12

57. A die is rolled twice. What is the probability that at least one 6 is obtained?

A. 11/36
B. 1/3
C. 1/36
D. 25/36
E. 1/6

58. Solve the system: x + y = 7 and x − y = 1.

A. x = 6, y = 1
B. x = 3, y = 4
C. x = 4, y = 3
D. x = 2, y = 5
E. x = 5, y = 2

59. The geometric sequence 2, 6, 18, ... has what 6th term?

A. 486
B. 243
C. 729
D. 1458
E. 162

60. A wire carrying a current of 3 A lies perpendicular to a uniform magnetic field of 0.2 T. What force acts on 0.5 m of the wire?

A. 0.3 N
B. 0.03 N
C. 30 N
D. 0.6 N
E. 3 N

정답 및 해설

번호	정답	해설
1	A	질병·식단·약물이 미생물을 바꿀 수 있음 → 역인과 가능성.
2	A	원인(미생물)을 먼저 주고 결과(증상)를 관찰 → 인과 방향 확인.
3	E	리타 레비-몬탈치니(NGF).
4	C	1948년.
5	E	"일부 항생제는 바이러스에 효과" 가 거짓 → 그 부정 "어떤 항생제도 효과 없음".
6	A	1.2 × 1.1 = 1.32 → 32%.
7	B	1인 1시간 3명 → 5×6×3 = 90.
8	D	피보나치: 8+13 = 21.
9	E	400 m를 더 달려야 함. 속도 5 − 4 = 1 m/s → 400 s.
10	C	미세소관.
11	A	유전 암호의 축중성.
12	D	예정된 세포사.
13	E	아세틸-CoA + 옥살로아세트산 → 시트르산.
14	A	딸은 아버지 X(정상) + 어머니 X 중 1/2 확률로 보인자 X.
15	D	물의 광분해.
16	B	소뇌.
17	D	글리세롤 + 지방산 2 + 인산기.
18	E	랑비에 결절 사이 도약.
19	D	혈소판.
20	B	$A^O \times B^O \rightarrow OO$ 1/4.
21	C	하수체 문맥계로 방출호르몬.
22	C	G–C 4×3 = 12, A–T 6×2 = 12 → 24.
23	C	중쇄 2 + 경쇄 2.
24	E	진한 소변 생성.
25	E	제1감수분열 전기.
26	D	활성부위 아미노산의 이온화 상태 변화.
27	A	터너 증후군.
28	E	식세포작용.
29	D	역전사효소.
30	E	중탄산염이 풍부한 액.
31	D	조면 ER 리보솜 → ER 내강 → 소포 → 골지 → 분비소포 → 세포막.
32	E	DNA·단백질 서열 유사성.
33	B	6 + 12 + 6 = 24.
34	B	N + 4(+1) = +1 → −3.
35	B	$[H^+]$ $10^{-4} \rightarrow 10^{-6}$ → pH 6.

번호	정답	해설
36	B	네온.
37	D	BF_3(비공유전자쌍 없음).
38	B	$CH_4 + 2O_2 \rightarrow 2\ \text{mol} \times 22.4 = 44.8\ \text{L}$.
39	C	에탄올.
40	C	0.05 mol / 0.5 L = 0.1 mol/L.
41	C	O–H를 가진 메탄올.
42	D	아세트산 이온의 가수분해 → OH^-.
43	D	$\text{rate} = k[A][B]^2$ → 3차.
44	A	$-CONH_2$.
45	D	방위 양자수 l.
46	C	$Na^+ + e^- \rightarrow Na$.
47	B	C_2H_6O: 다이메틸 에터.
48	B	$v^2 = 2as \rightarrow 900 = 150a \rightarrow a = 6$.
49	C	$250 - mgh(200) = 50\ \text{J}$.
50	A	$P = V^2/R = 0.75\ \text{W} \times 20 = 15\ \text{J}$.
51	C	$T \propto \sqrt{L}$ → 2배.
52	E	800/1000 = 0.8.
53	A	무게 = mg = 30 N(자유낙하에서도 중력은 작용).
54	B	1/0.02 = 50 Hz.
55	B	log 100 = 2.
56	E	5-12-13.
57	A	$1 - (5/6)^2 = 11/36$.
58	C	x = 4, y = 3.
59	A	$2 \times 3^5 = 486$.
60	A	$F = BIL = 0.2 \times 3 \times 0.5 = 0.3\ \text{N}$.